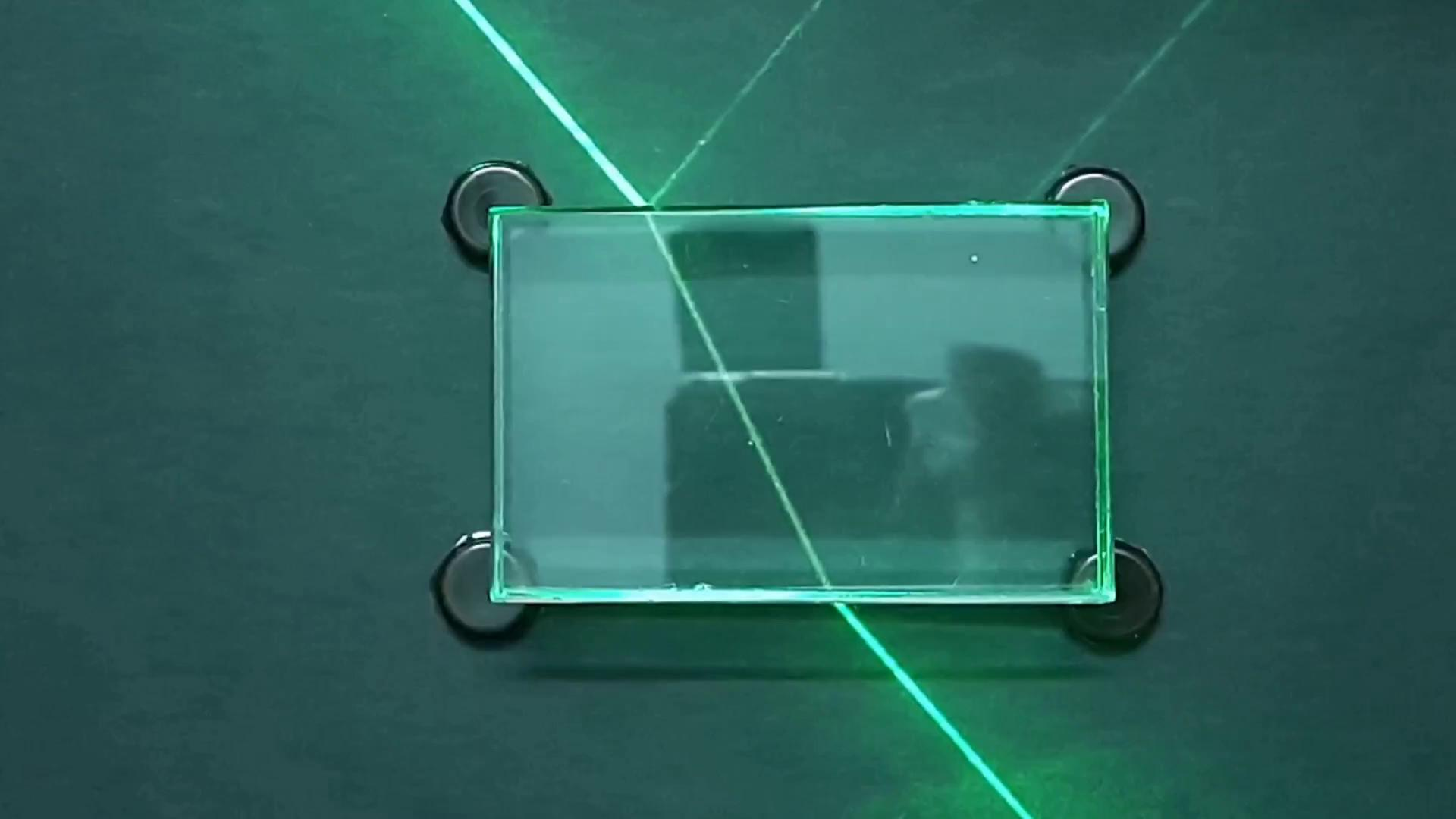
CHAPTER 1 : LIGHT REFLECTION AND REFRACTION

Learning Objectives :

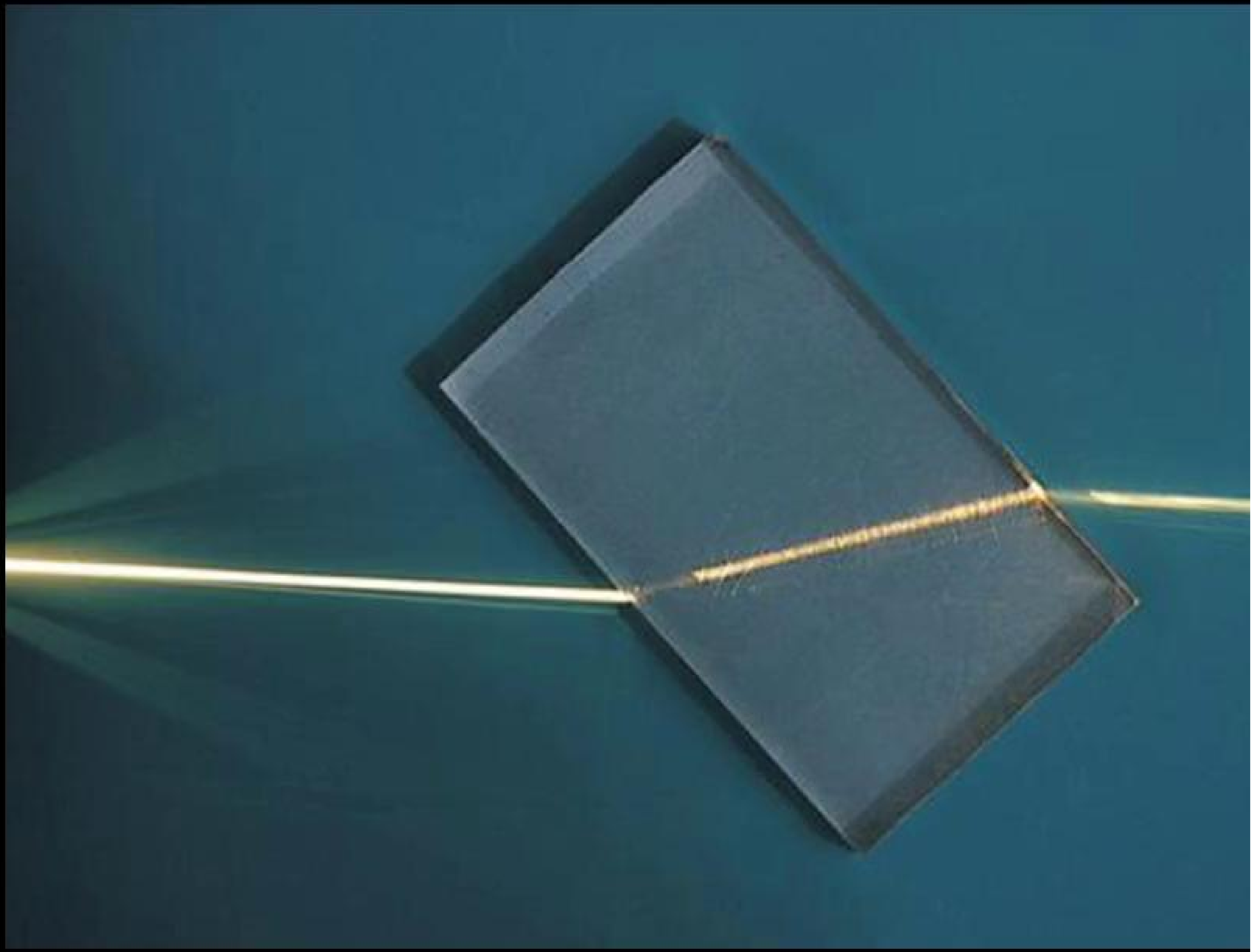
- Introduction
- Reflection And Laws of reflection
- Spherical Mirrors
- Image formation by Spherical Mirrors
- Ray Diagrams
- Uses of Concave and Convex Mirrors
- Sign Convention
- Mirror Formula and Magnification
- Refraction of Light
- Refraction through a Rectangular Glass Slab
- Laws of Refraction
- The Refractive Index
- Refraction by Spherical Lenses
- Image Formation by Lenses & Ray Diagrams
- Lens Formula & Magnification
- Power of a Lens



Refraction of Light

- When traveling obliquely from one medium to another, the direction of propagation of light in the second medium changes. This phenomenon is known as refraction of light.

OR



- The bending of light when it enters obliquely from one transparent medium to another.

- The major cause of refraction to occur is the change in the speed of light in different mediums

- Speed of light is maximum in vacuum : $c = 3 \times 10^8$ m/s

- Fermat's Principle : A ray of light going from point A to point B always takes the route of least time.

Refraction of Light

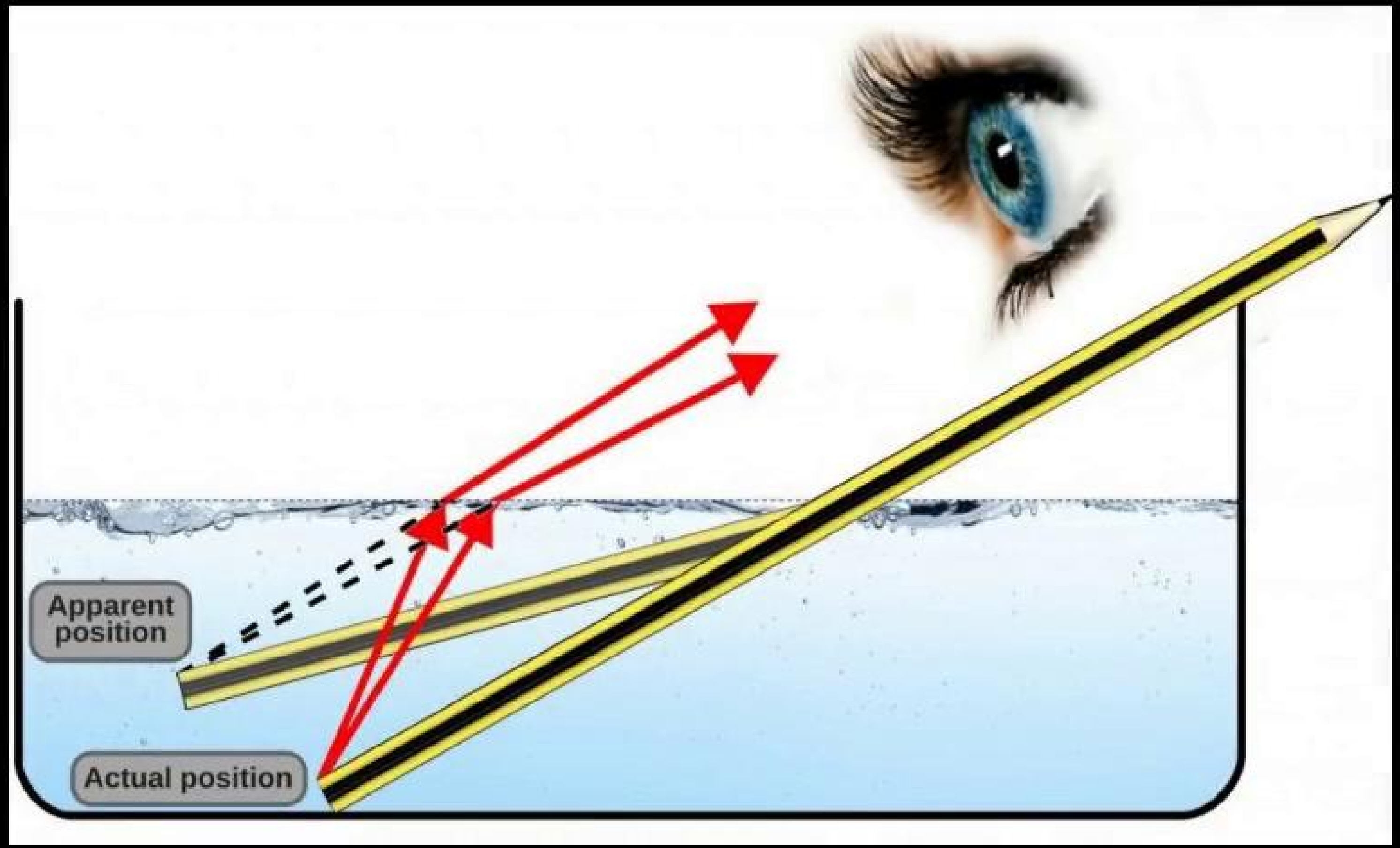
Some Examples of Refraction :

- (1) The bottom of swimming pool appears higher

Refraction of Light

Some Examples of Refraction :

- (1) The bottom of swimming pool appears higher
- (2) The pencil partially immersed in water appears to be bent at the interface



Refraction of Light

Some Examples of Refraction :

- (1) The bottom of swimming pool appears higher
- (2) The pencil partially immersed in water appears to be bent at the interface
- (3) Lemons placed in a glass tumbler appears bigger

Refraction of Light

Some Examples of Refraction :

- (1) The bottom of swimming pool appears higher

- (2) The pencil partially immersed in water appears to be bent at the interface

- (3) Lemons placed in a glass tumbler appears bigger

- (4) Letters of a book appear to be raised when seen through a glass slab.

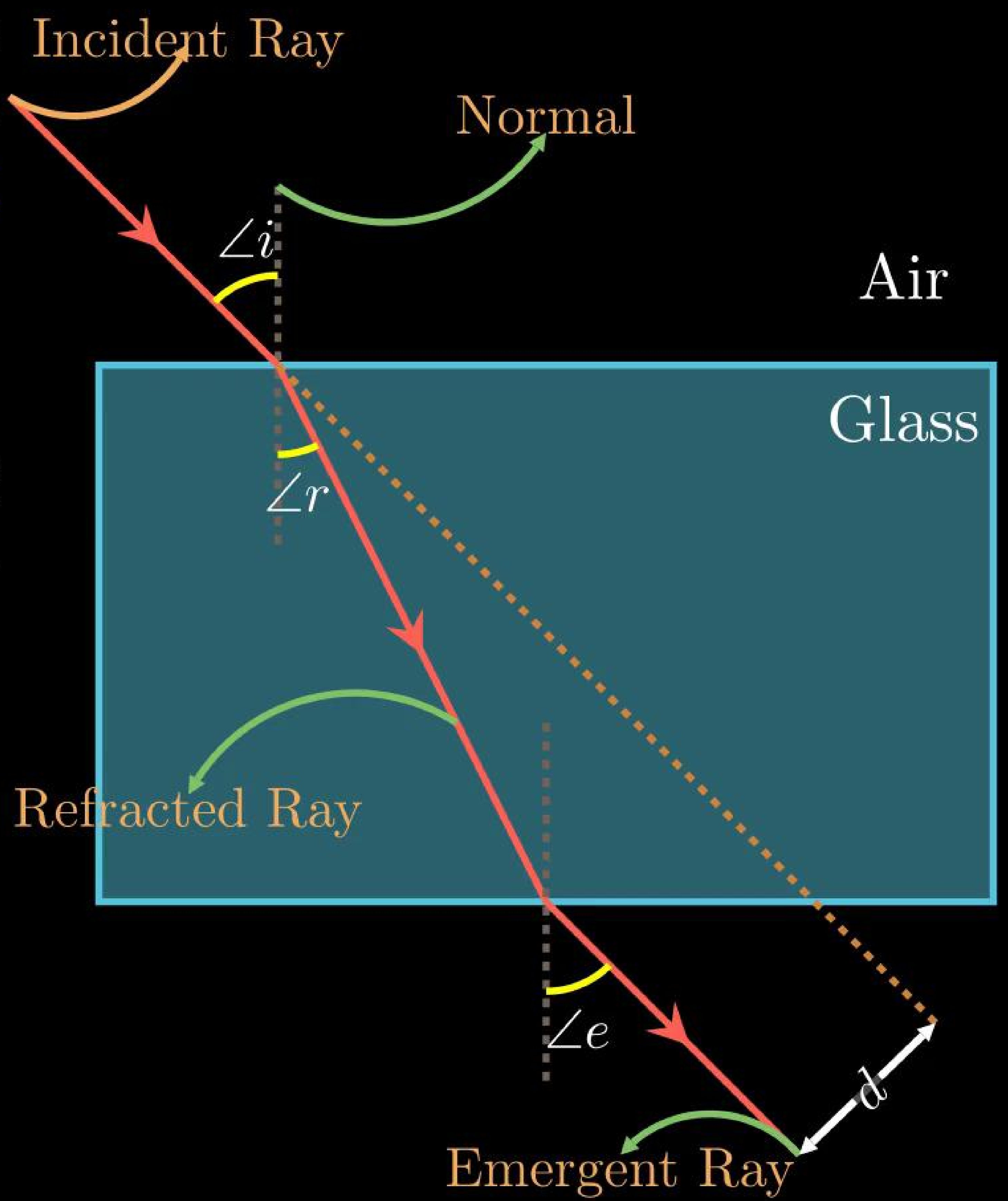
Refraction through a Rectangular Glass Slab

- The extent of bending of a ray of light at the opposite parallel faces of a rectangular glass slab is equal and opposite, so the ray emerges parallel to incident ray.
- For glass slab : $\angle i = \angle e$
- The perpendicular distance between direction of incident ray and emergent ray is called lateral displacement (d)
- Lateral displacement (d) depends on -

(1) Refractive index of the material of the slab.

(2) Thickness of the slab.

(3) Angle of incidence.



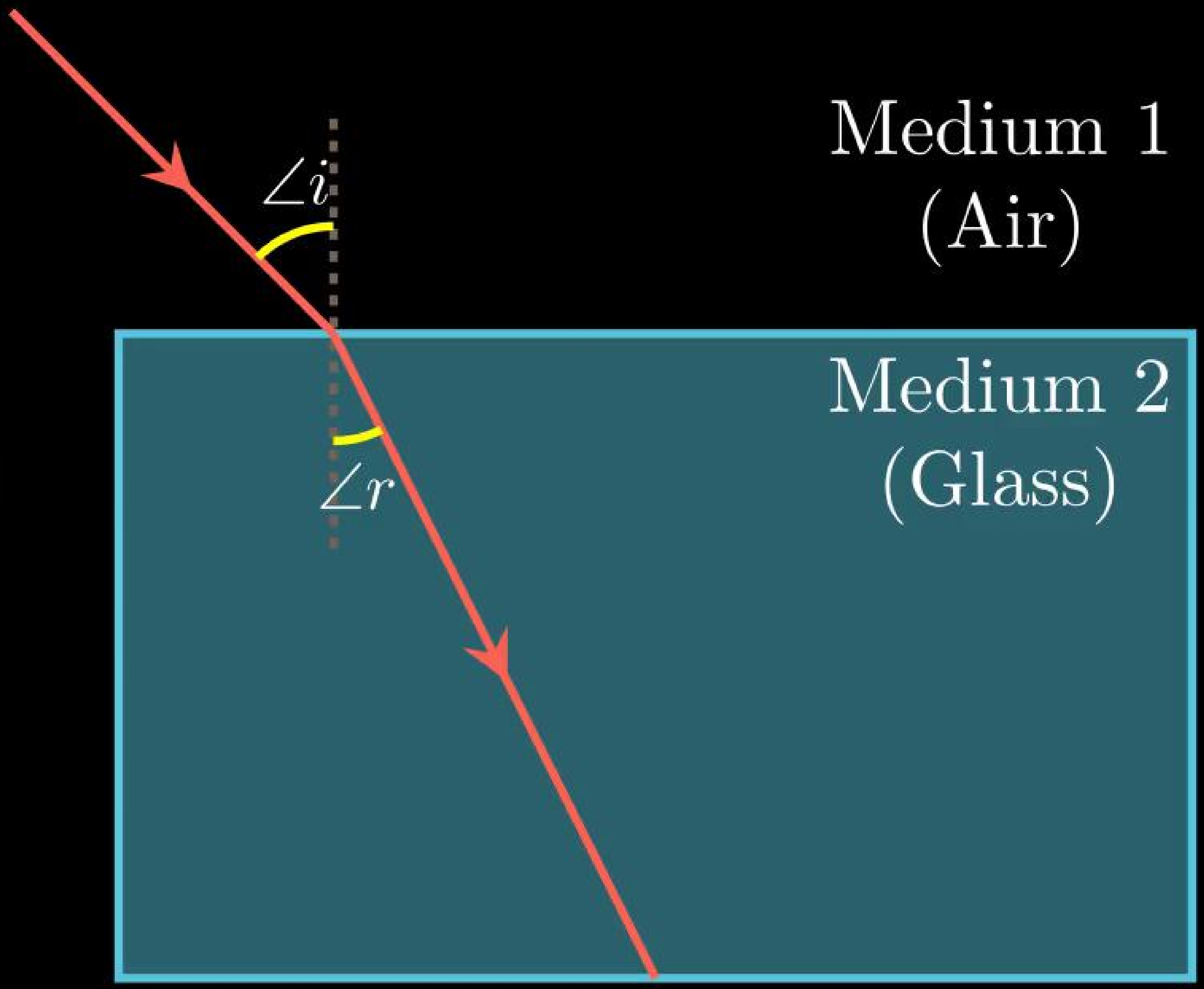
Laws of Refraction

- (1) The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane.

- (2) Snell's law : The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant.

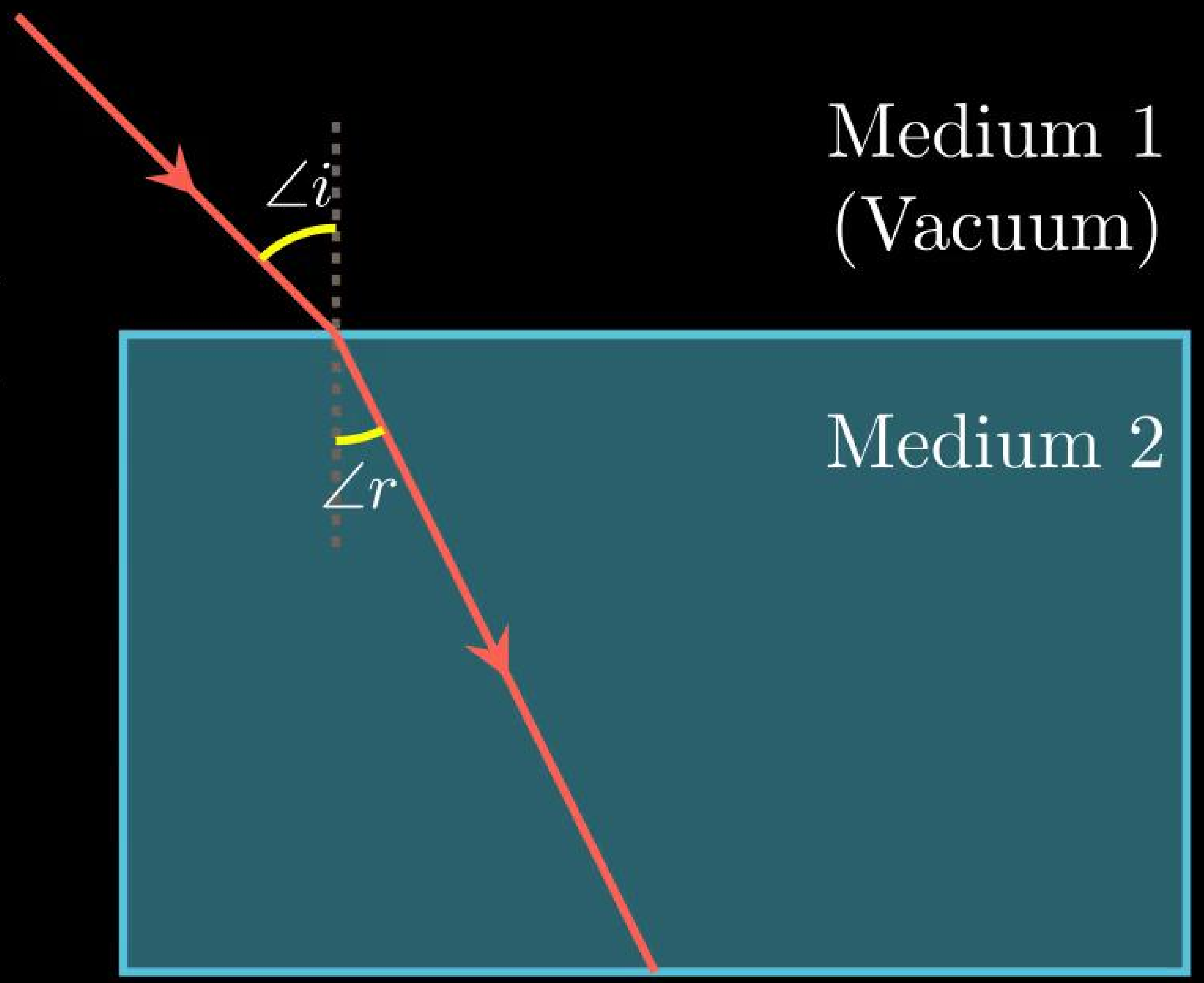
$$\boxed{\frac{\sin(\angle i)}{\sin(\angle r)} = \text{Constant} = n_{21}}$$

This constant value is called the refractive index (n_{21}) of the second medium with respect to the first.



The Refractive Index

- Different medium have different abilities to bend or refract light. This bending ability of a medium is known as the refractive index.



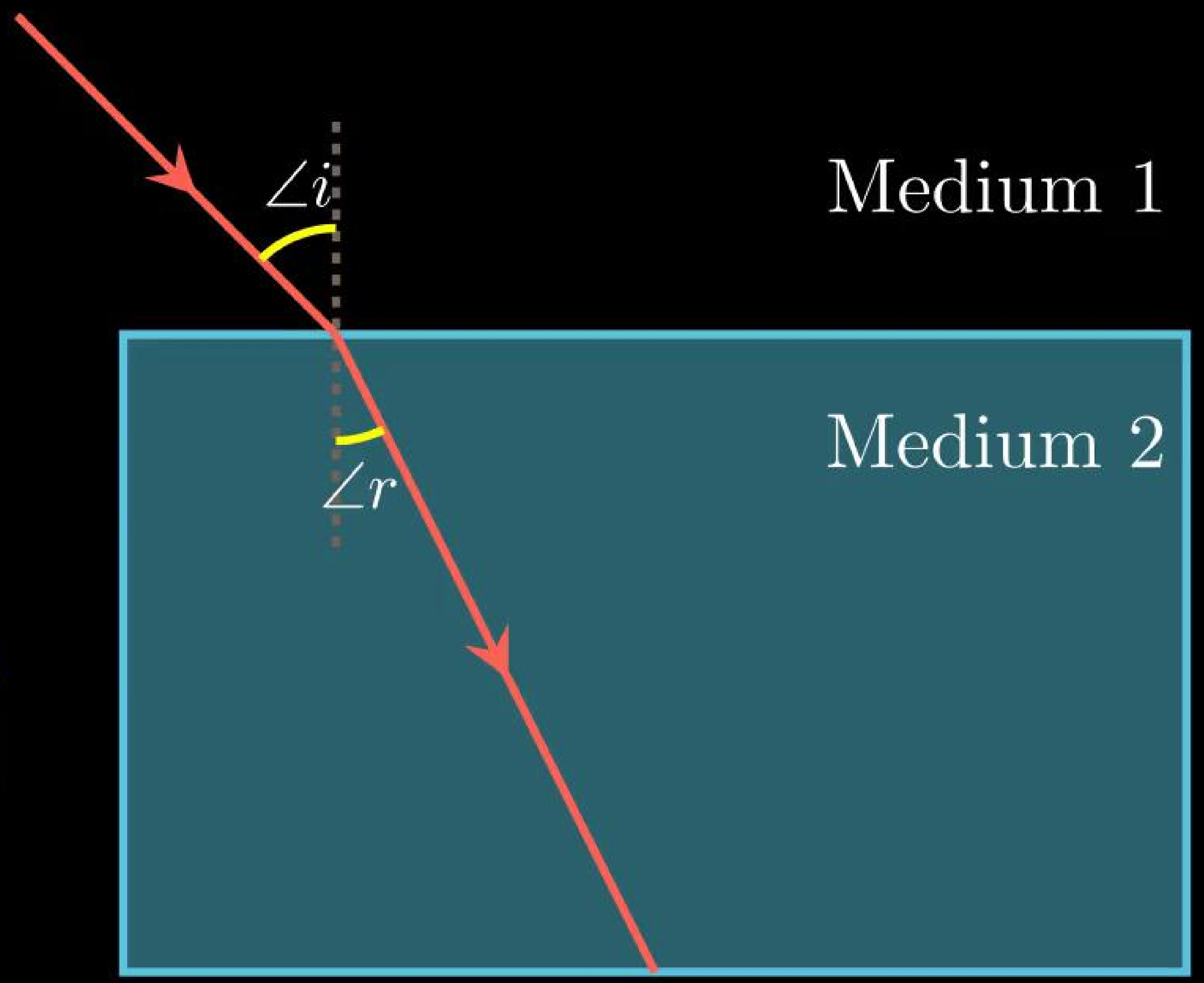
- Types of Refractive index :
  (1) Absolute refractive index (n)

  (2) Relative Refractive index (n_{21})

- (1) Absolute Refractive index : Absolute refractive index or simply **refractive index of Medium 2** (n_2) is defined as the ratio of speed of light in vacuum (c) (i.e., Medium 1) to the speed of light in the Medium 2 (v_2)

$$\boxed{n_2 = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium 2}} = \frac{c}{v_2}}$$

The Refractive Index



- (2) Relative Refractive index : Relative refractive index of medium 2 with respect to medium 1 (n_{21}) is defined as the ratio of speed of light in medium 1 (v_1) to the speed of light in the Medium 2 (v_2)

$$\boxed{n_{21} = \frac{\text{speed of light in medium 1}}{\text{speed of light in medium 2}} = \frac{v_1}{v_2} = \frac{n_2}{n_1} = \frac{\sin(\angle i)}{\sin(\angle r)}}$$

The Refractive Index

Material Medium	Refractive Index	Material Medium	Refractive Index
Air	1.0003	Crown glass	1.52
Ice	1.31	Canada Balsam	1.53
Water	1.33	Rock salt	1.54
Alcohol	1.36	Carbon disulphide	1.63
Kerosene	1.44	Dense flint glass	1.65
Fused quartz	1.46	Ruby	1.71
Turpentine oil	1.47	Sapphire	1.77
Benzene	1.50	Diamond	2.42

Table: Absolute refractive index of some material media

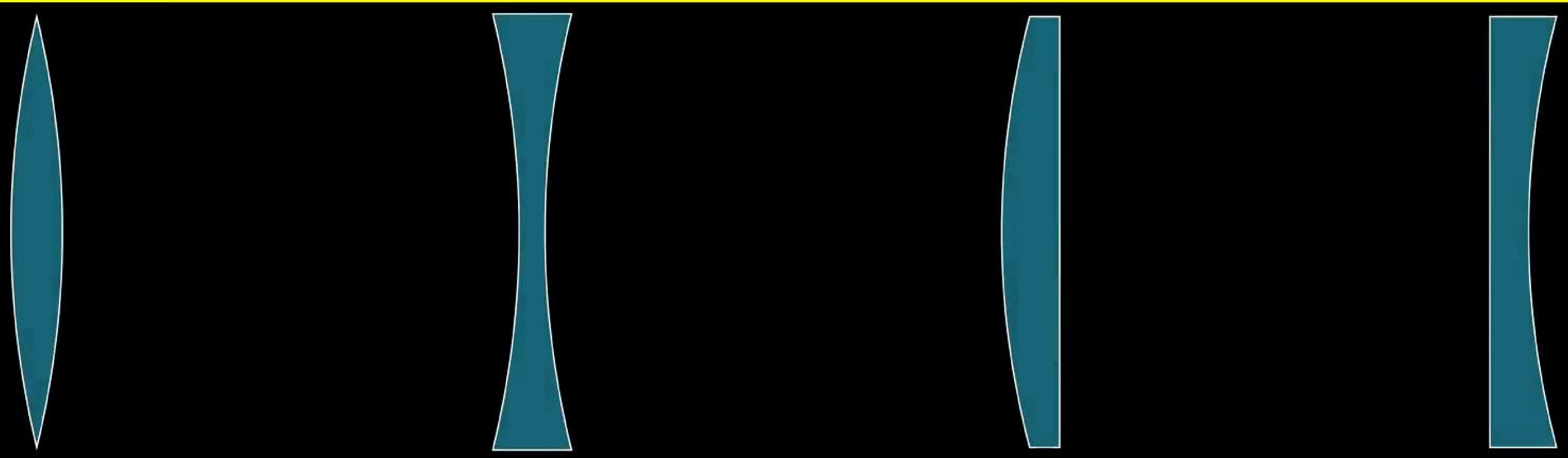
The Refractive Index

- Optical Density : The ability of a medium to refract light is also expressed in terms of its optical density.
- Optical density is not the same as mass density.
- Denser Medium : In comparing two media, the one with the larger refractive index is optically denser medium than the other.
- Rarer Medium : The other medium of lower refractive index is optically rarer.
- The speed of light is higher in a rarer medium than a denser medium
- Thus, a ray of light traveling from a rarer medium to a denser medium slows down and bends **towards the normal**.
- When it travels from a denser medium to a rarer medium, it speeds up and bends **away from the normal**.

Refraction by Spherical Lenses

- SPHERICAL LENS : A spherical lens is a transparent material bounded by two surfaces one or both of which are spherical.

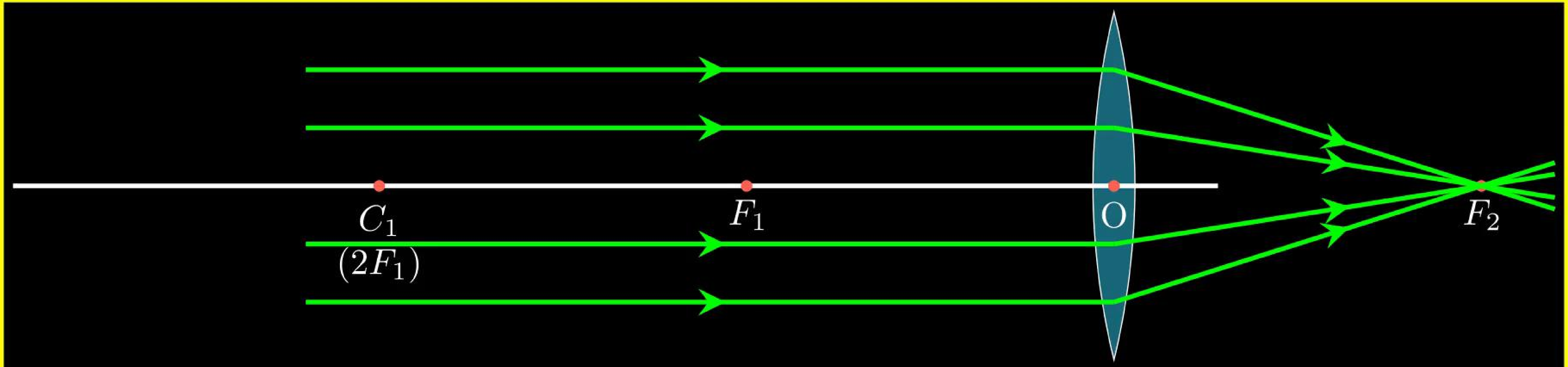
- A lens is bound by at least one spherical surface. In such lenses, the other surface would be plane. Spherical lenses are of four main types



Double Convex Lens (Or Convex Lens) | Double Concave Lens (Or Concave Lens) | Plano-Convex Lens | Plano-Concave Lens

Refraction by Spherical Lenses

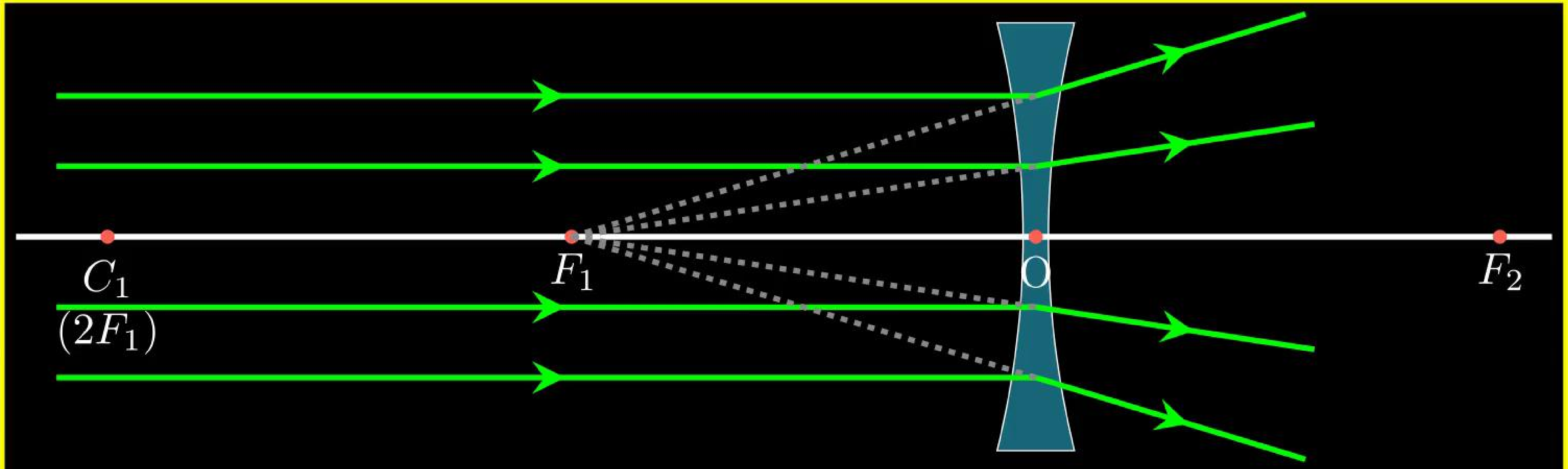
- Convex lens : A double convex lens is bounded by two spherical surfaces, **curved outwards (bulging outwards)**.
- It is **thicker in the middle** and **thinner at the edges**.
- Convex lens converges light rays. So, it is also called **converging lenses**.



Converging action of a convex lens.

Refraction by Spherical Lenses

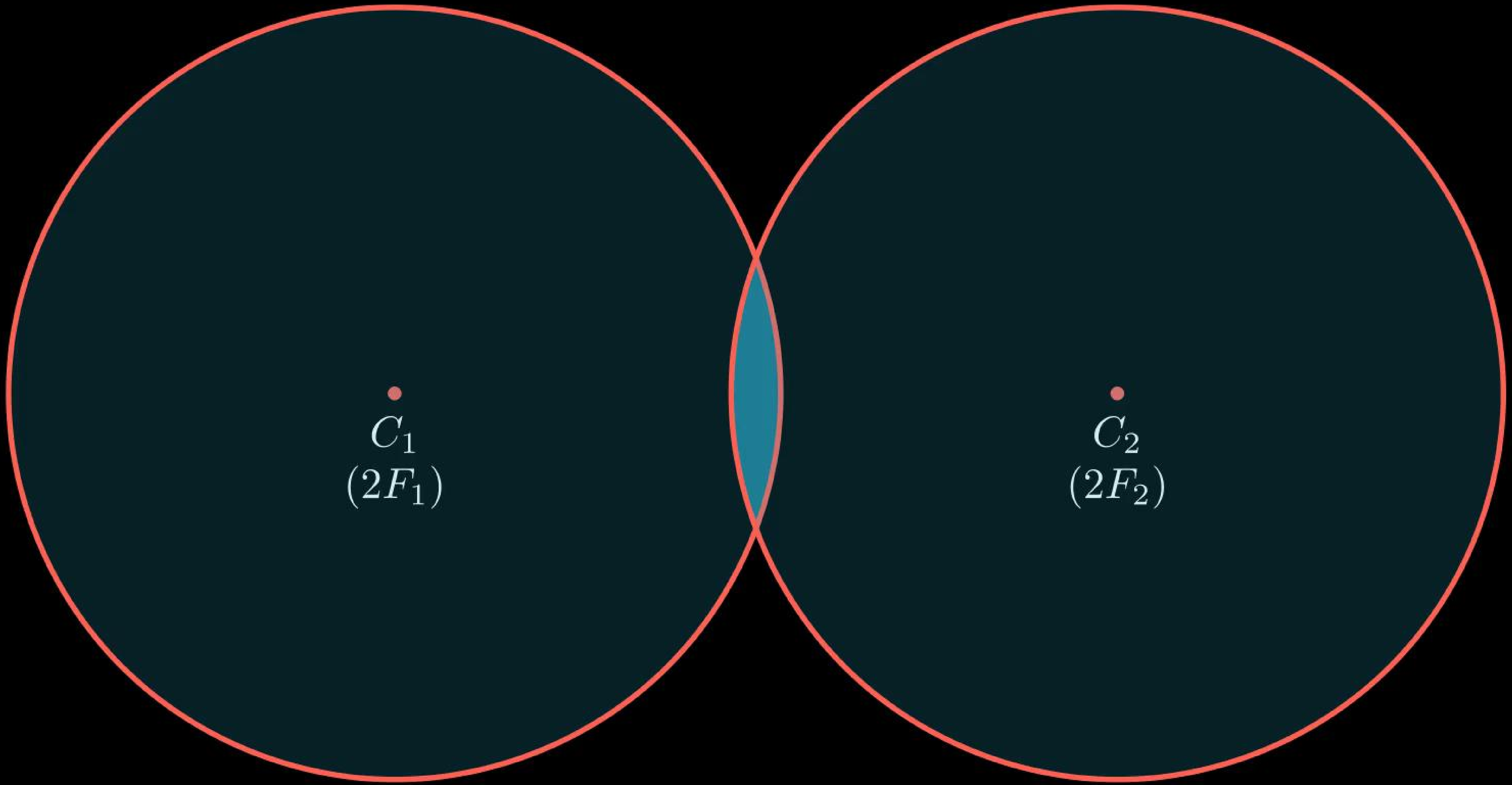
- Concave lens : A double convex lens is bounded by two spherical surfaces, **curved inwards.**.
- It is **thinner in the middle** and **thicker at the edges**.
- Convex lens diverges light rays. So, it is also called **diverging lenses**.



Diverging action of a concave lens.

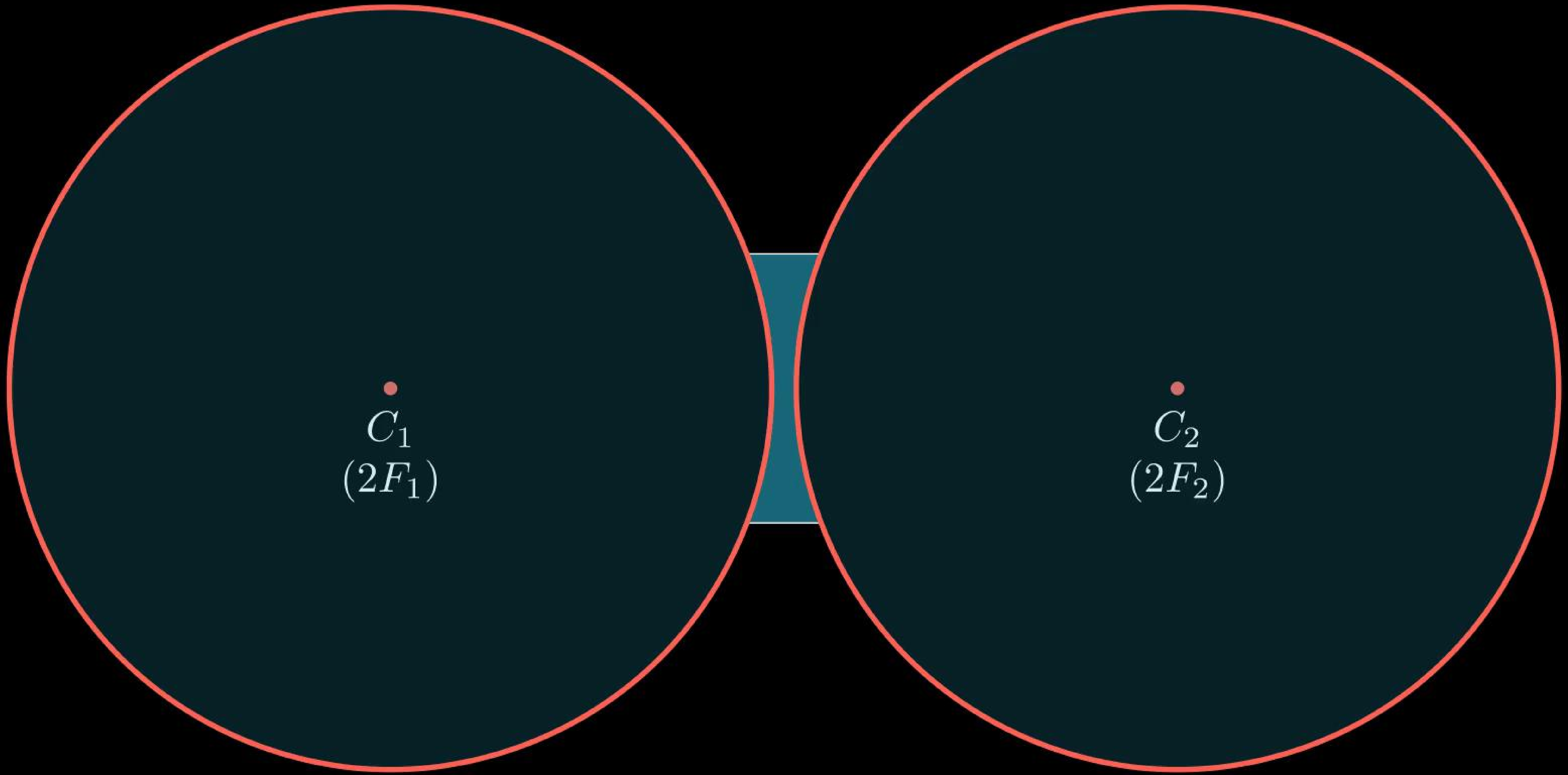
Important Terms Related to Spherical Lenses:

- (1) Centre of Curvature (C):



Important Terms Related to Spherical Lenses:

- (1) Centre of Curvature (C): The centres of the spheres that the spherical lens was a part of. A spherical lens has two centres of curvatures ($C_1,\ C_2$).

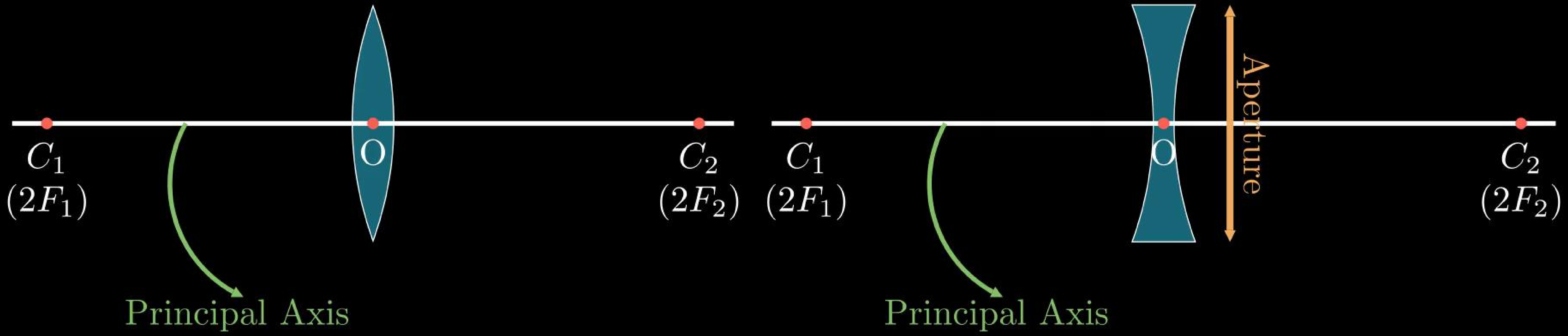


Important Terms Related to Spherical Lenses:

- (2) Principal Axis: An imaginary straight line passing through the two centres of curvature of a lens is called its principal axis.

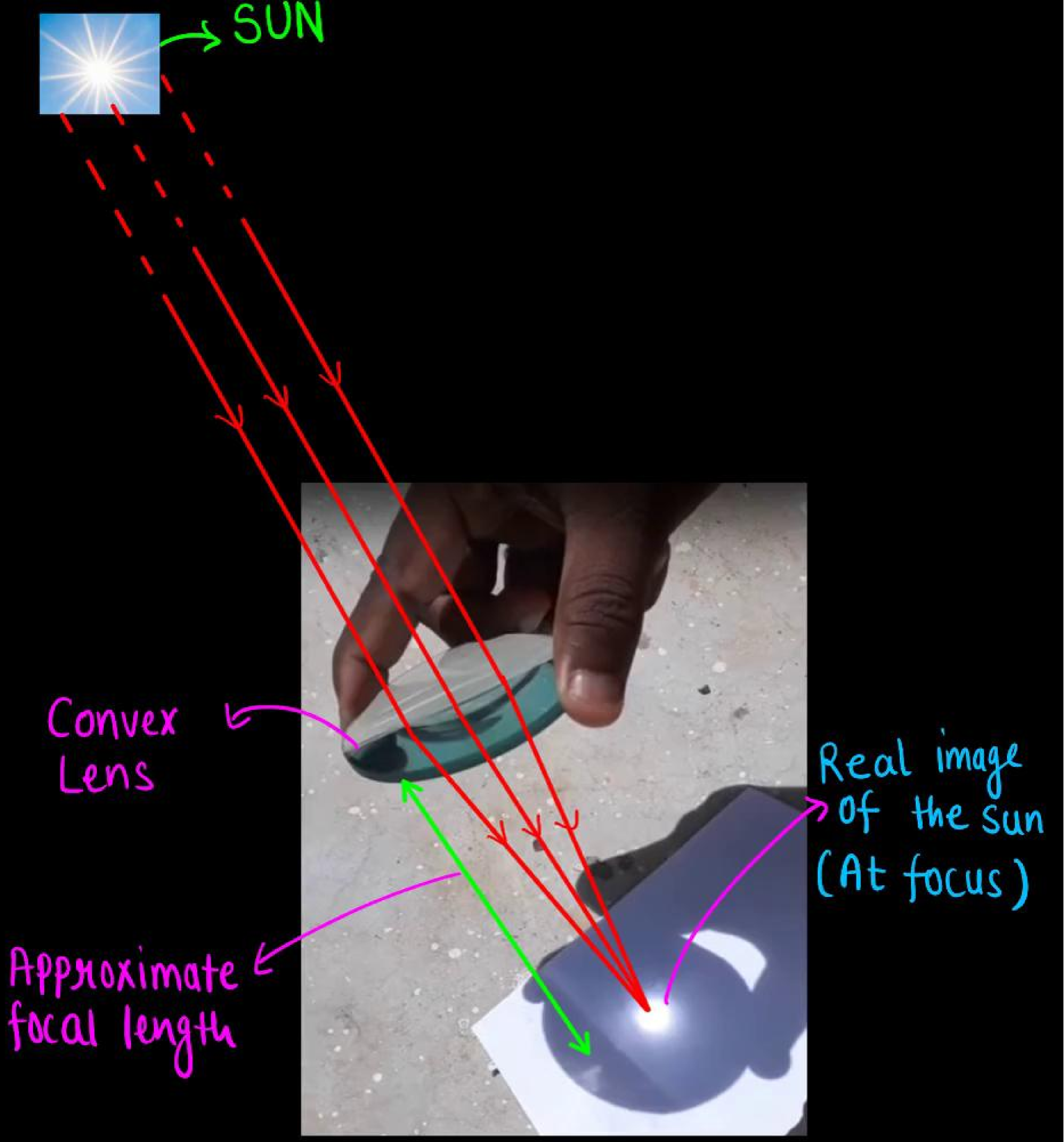
- (3) Optical Centre (O): The central point of a lens is its optical centre. A ray of light through the optical centre of a lens passes without suffering any deviation.

- (4) Aperture : The effective diameter of the circular outline of a spherical lens is called its aperture. [Thin lenses $\rightarrow$ small aperture$(< R)$ and $OC_1 = OC_2 = R$]

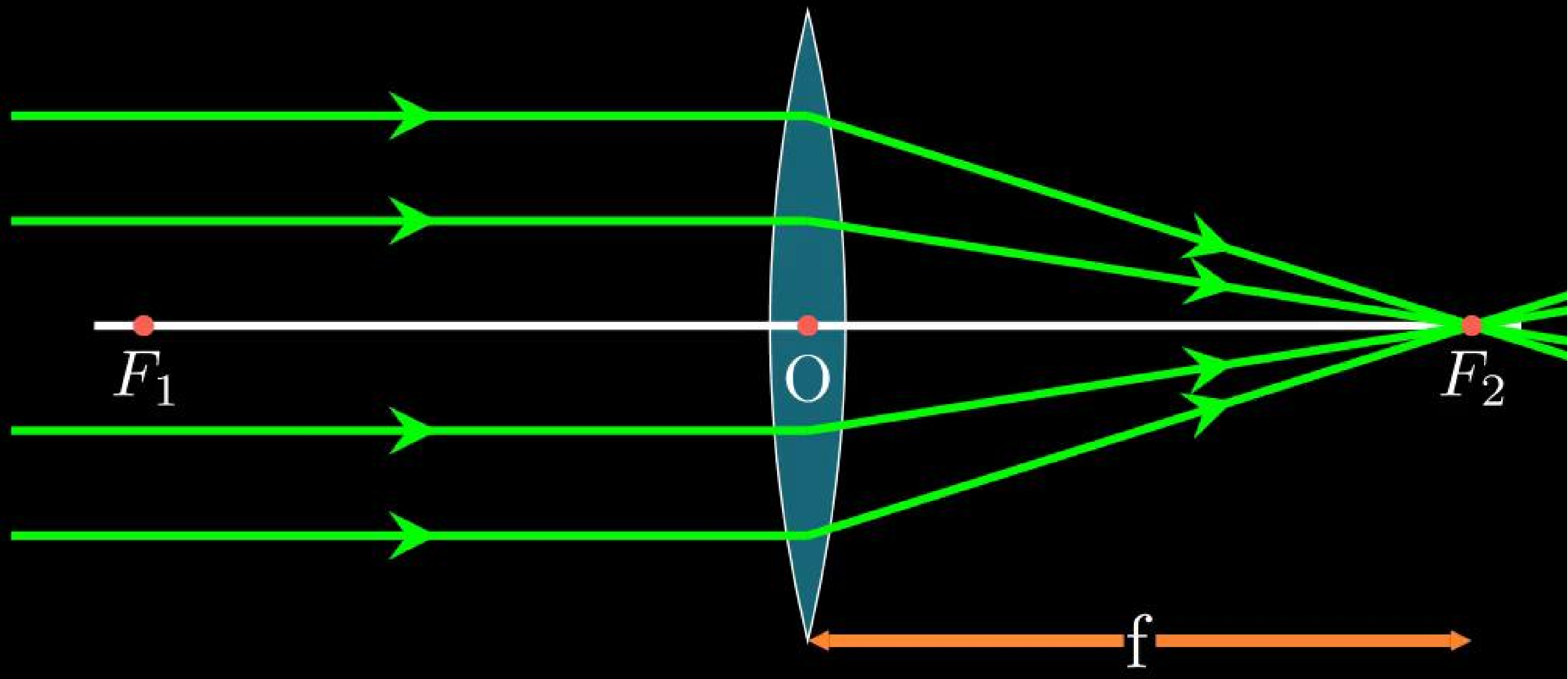


Activity 10.11:

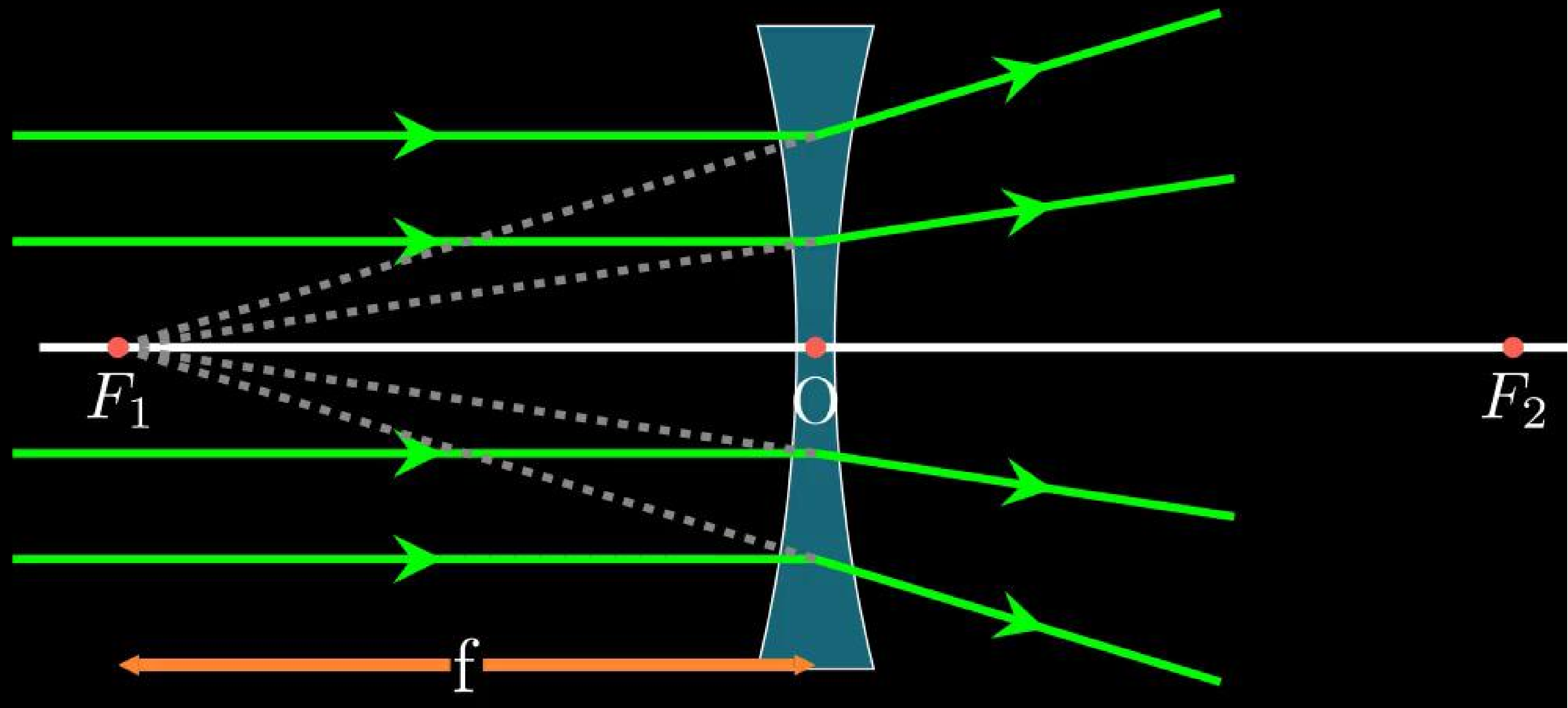
- Hold a convex lens in your hand. Direct it towards the Sun.
- Focus the light from the Sun on a sheet of paper. Obtain a sharp bright image of the Sun.
- Hold the paper and the lens in the same position for a while. Keep observing the paper.
- We observe that the paper begins to burn producing smoke.
- Parallel sun rays were converged by the lens at the sharp bright spot formed on the paper. The concentration of the sunlight at a point generated heat. This caused the paper to burn.



- (5) Principal Focus of Convex Lens : When several rays of light parallel to the principal axis are falling on a convex lens. These rays, after refraction from the lens, are converging to a point on the principal axis. This point on the principal axis is called the principal focus of the len



- (6) Principal Focus of Concave Lens : When several rays of light parallel to the principal axis are falling on a convex lens. These rays, after refraction from the lens, are appearing to diverge from a point on the principal axis. This point on the principal axis is called the principal focus of the lens



- Focal Length (f) : distance of the principal focus from the optical centre.

Image Formation by Lenses and Ray Diagrams

Activity 10.12 : Image formation by a convex lens for different positions of the object

	Position of the Object	Position of the Image	Size of the Image	Nature of the Image
(1)	At infinity	At focus F_2	Highly diminished, point-sized	Real and inverted
(2)	Beyond C_1 $(2F_1)$	Between F_2 and $C_2(2F_2)$	Diminished	Real and inverted
(3)	At C_1 $(2F_1)$	At $C_2(2F_2)$	Same size	Real and inverted
(4)	Between C_1 $(2F_1)$ and F_1	Beyond C_2 $(2F_2)$	Enlarged	Real and inverted
(5)	At F_1	At infinity	Highly enlarged	Real and inverted
(6)	Between O and F_1	On the same side of the lens as the object	Enlarged	Virtual and erect

Image Formation by Lenses and Ray Diagrams

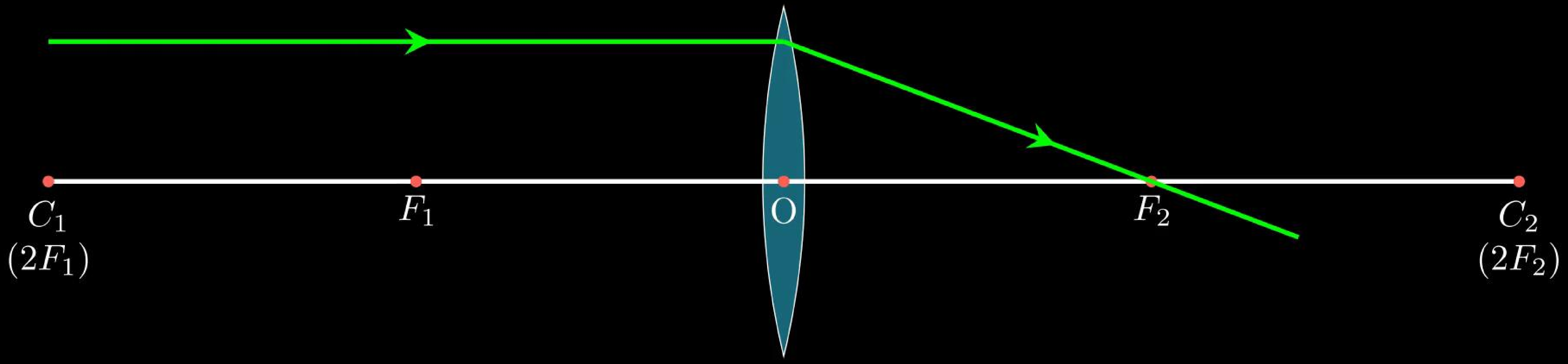
Activity 10.13 : Image formation by a concave lens for different positions of the object

	Position of the Object	Position of the Image	Size of the Image	Nature of the Image
(1)	At infinity	At focus F_1	Highly diminished, point-sized	Virtual and erect
(2)	Between infinity and optical centre O of the lens	Between focus F_1 and optical centre O	Diminished	Virtual and erect

Note: A concave lens will always give a virtual, erect and diminished image, irrespective of the position of the object.

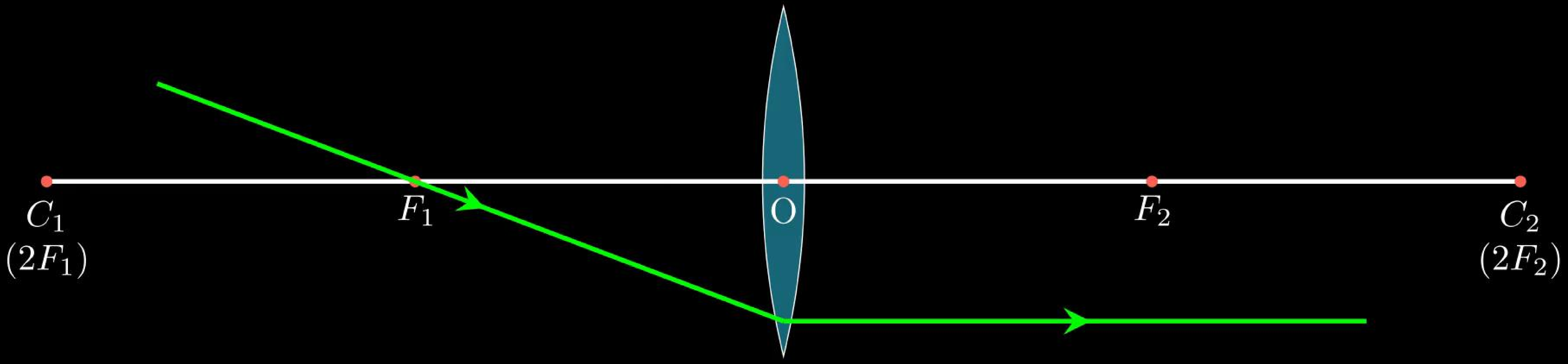
Ray Diagram for Convex Lens

(1) A ray of light parallel to the principal axis of a convex lens, passes through the principal focus on the other side of the lens.



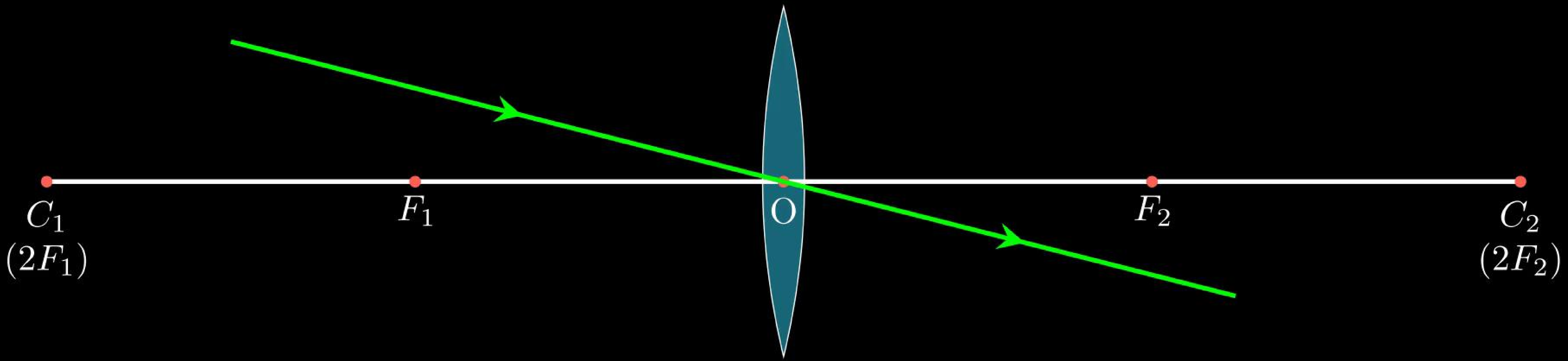
Ray Diagram for Convex Lens

(2) A ray of light passing through a principal focus, after refraction from a convex lens, will emerge parallel to the principal axis.

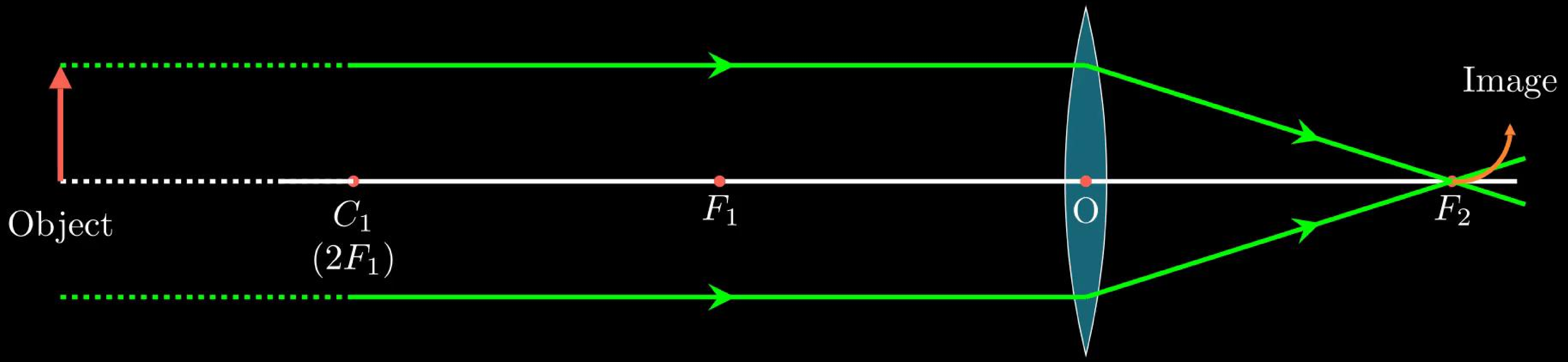


Ray Diagram for Convex Lens

(3) A ray of light passing through the optical centre of a lens will emerge without any deviation.

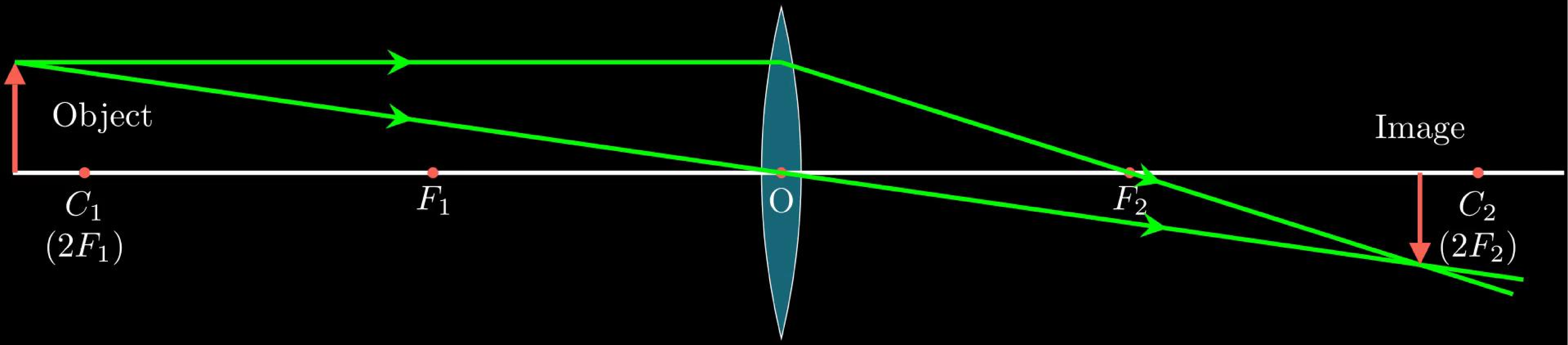


Position of Object (i): At infinity



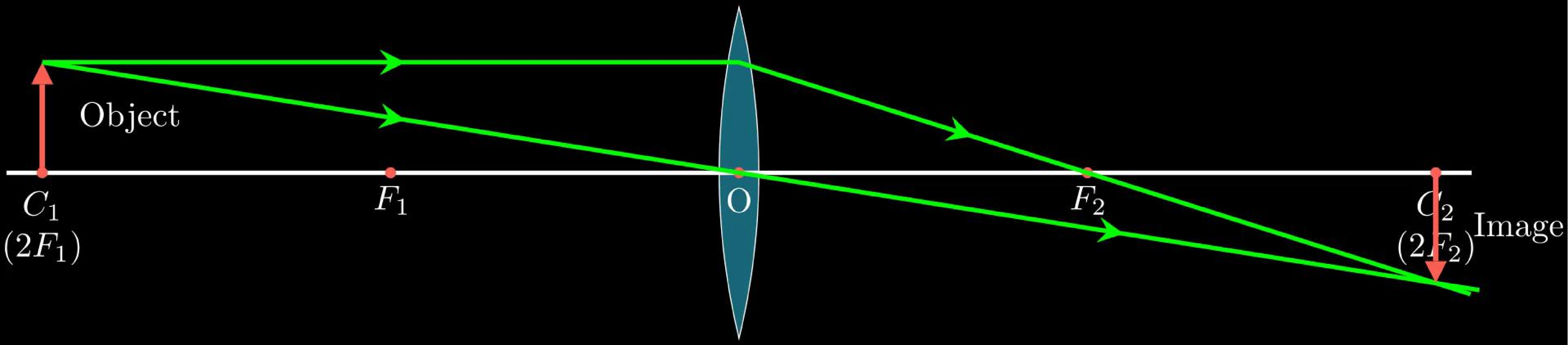
Position of the Image	Size of the Image	Nature of the Image
At the focus (F_2)	Highly diminished, point-sized	Real and inverted

Position of Object (ii): Beyond C_1 $(2F_1)$



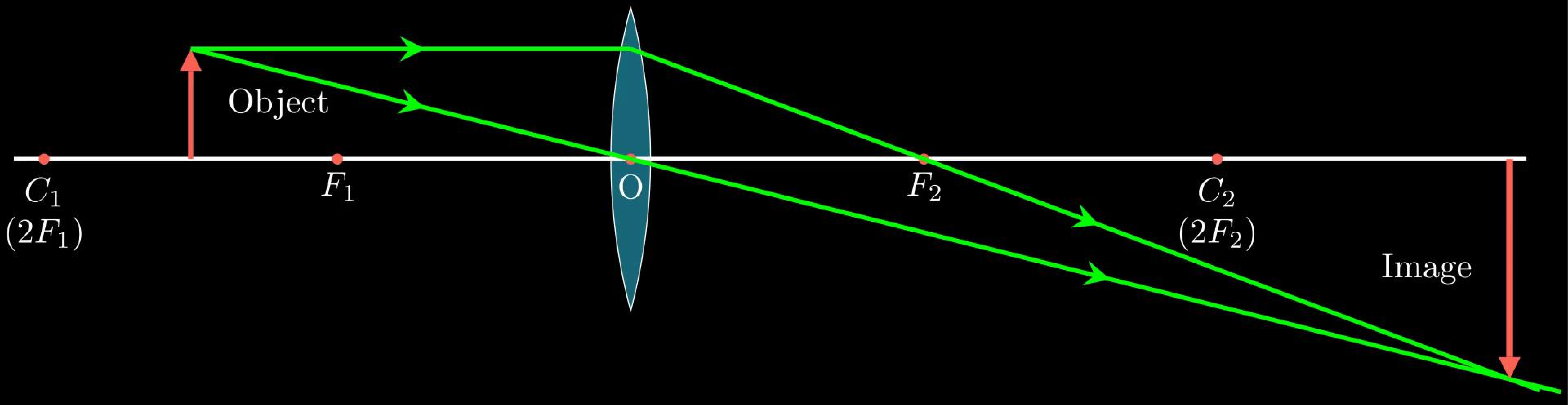
Position of the Image	Size of the Image	Nature of the Image
Between F_2 and $C_2(2F_2)$	Diminished	Real and inverted

Position of Object (iii): At C_1 $(2F_1)$



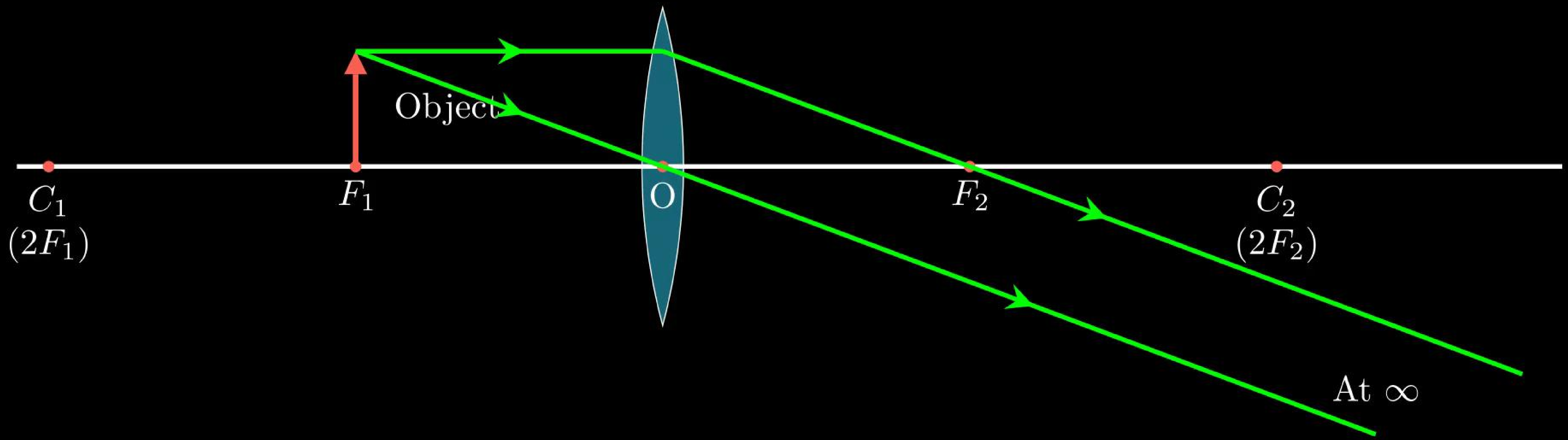
Position of the Image	Size of the Image	Nature of the Image
At $C_2(2F_2)$	Same Size	Real and inverted

Position of Object (iv): Between C_1 and F_1



Position of the Image	Size of the Image	Nature of the Image
Beyond C_2 $(2F_2)$	Enlarged	Real and inverted

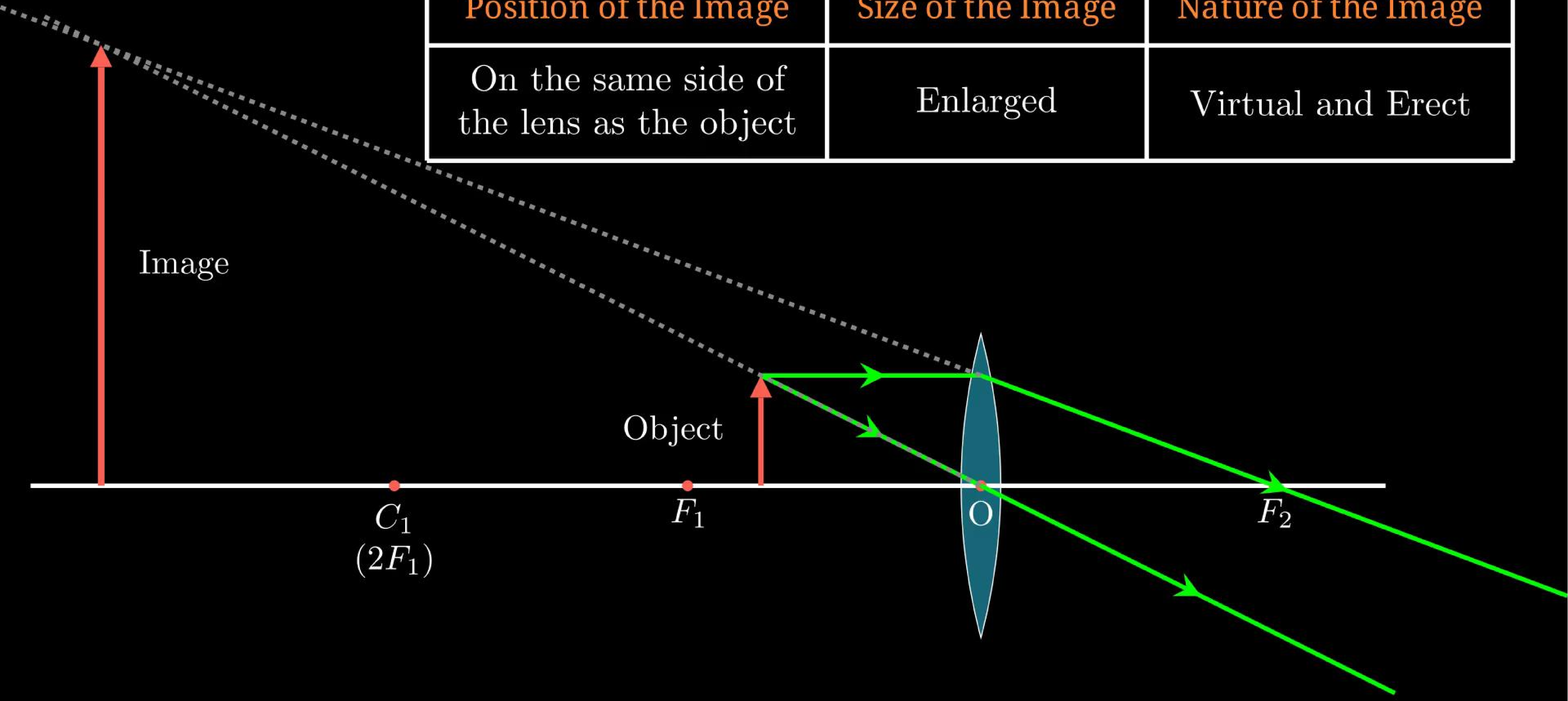
Position of Object (v): At F_1



Position of the Image	Size of the Image	Nature of the Image
At infinity (∞)	Highly Magnified (Highly Enlarged)	Real and inverted

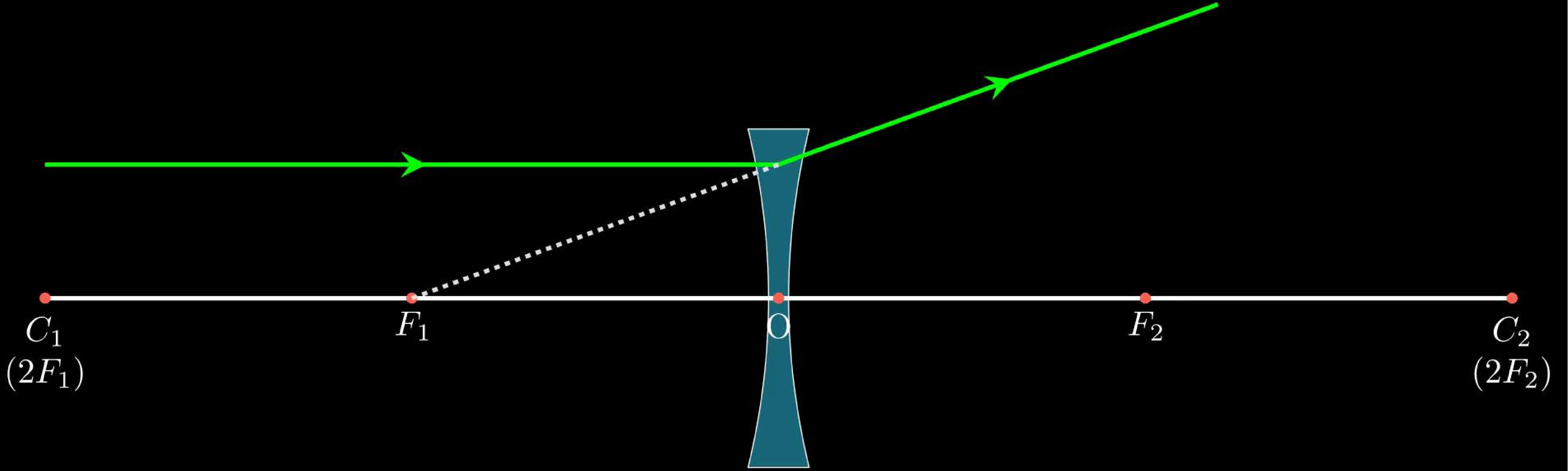
Position of Object (vi): Between F_1 and O

Position of the Image	Size of the Image	Nature of the Image
On the same side of the lens as the object	Enlarged	Virtual and Erect



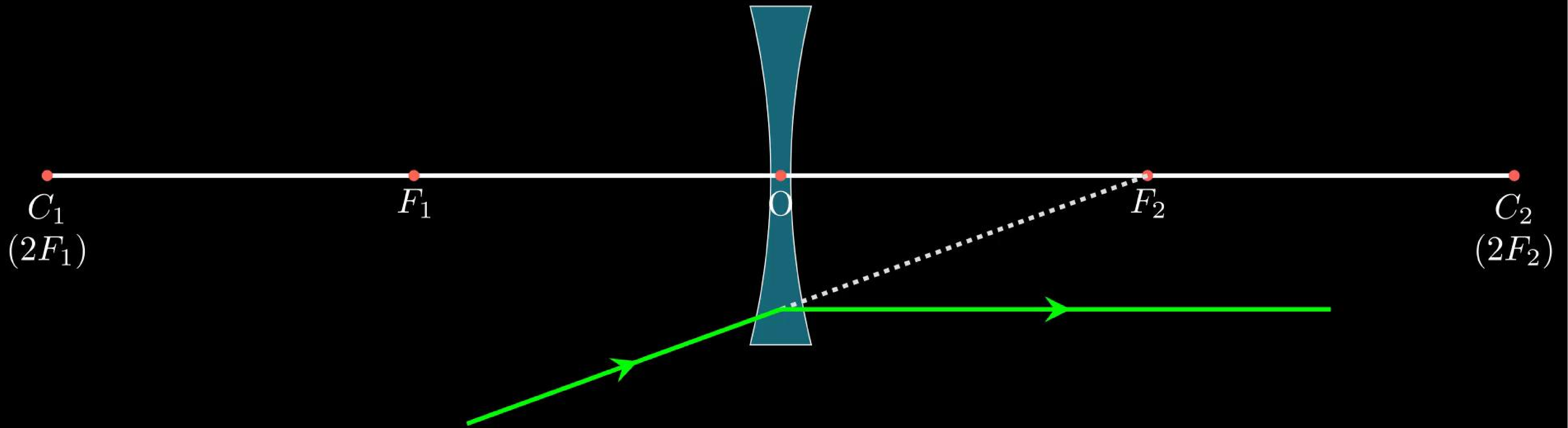
Ray Diagram for Concave Lens

(1) A ray of light parallel to the principal axis of a concave lens,after refraction the ray appears to diverge from the principal focus located on the same side of the lens



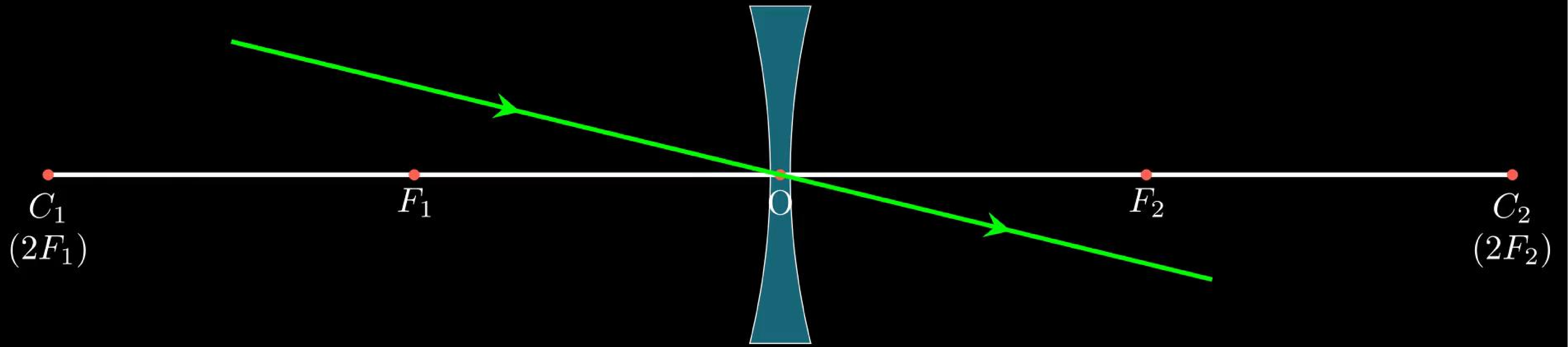
Ray Diagram for Concave Lens

(2) A ray of light appearing to meet at the principal focus of a concave lens, after refraction from a concave lens, will emerge parallel to the principal axis.

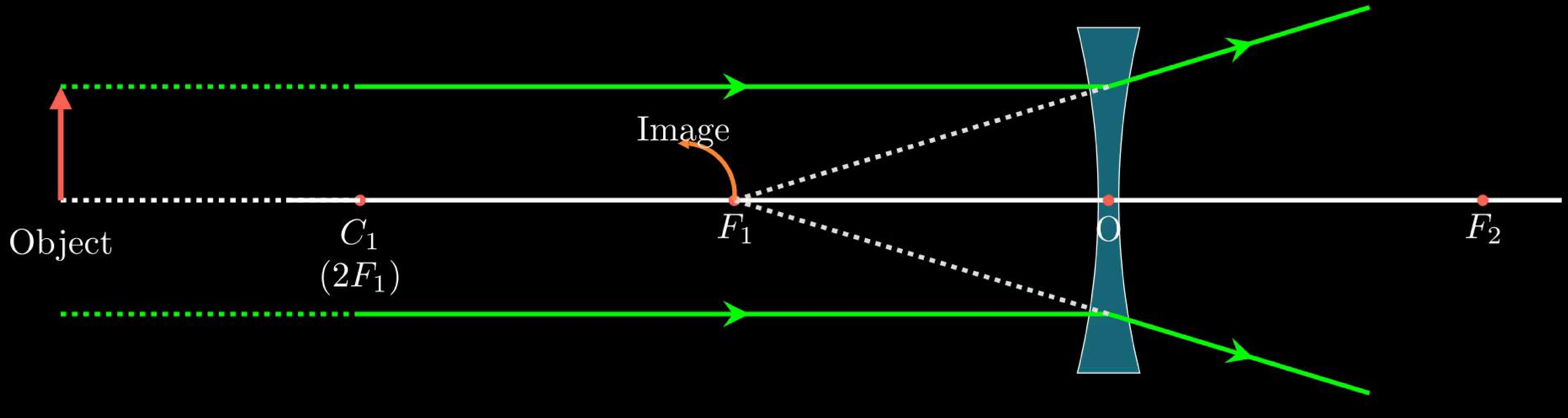


Ray Diagram for Concave Lens

(3) A ray of light passing through the optical centre of a lens will emerge without any deviation.

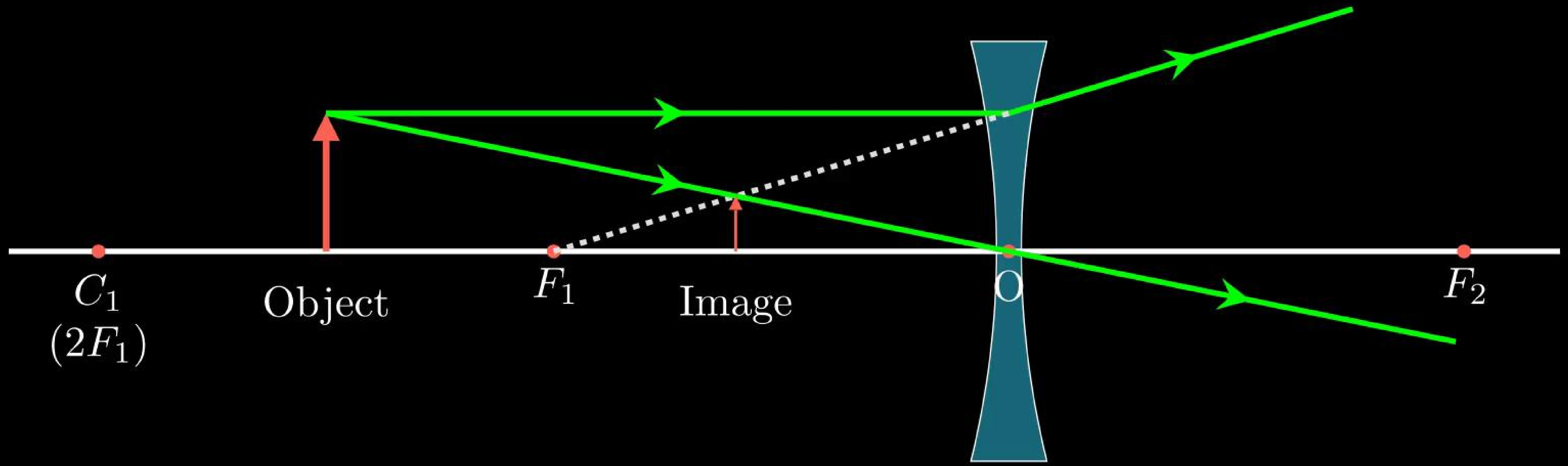


Position of Object (i): At infinity



Position of the Image	Size of the Image	Nature of the Image
At the focus (F_1)	Highly diminished, point-sized	Virtual and Erect

Position of Object (ii): Between infinity and optical centre O of the lens



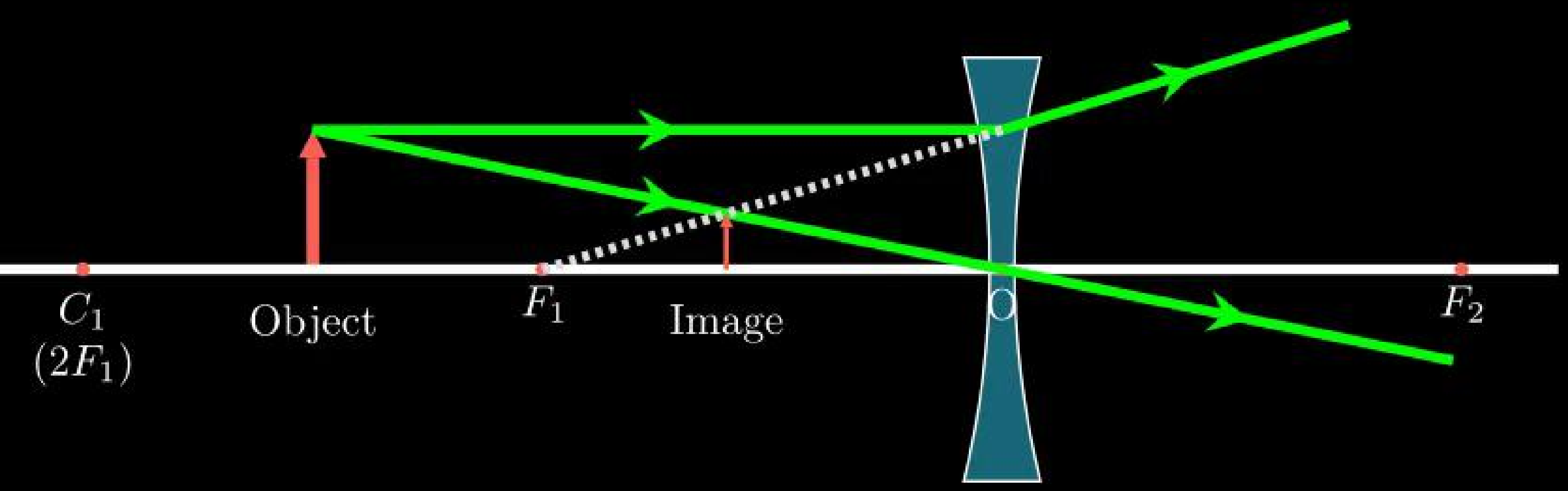
Position of the Image	Size of the Image	Nature of the Image
Between focus (F_1) and optical centre O	Diminished	Virtual and Erect

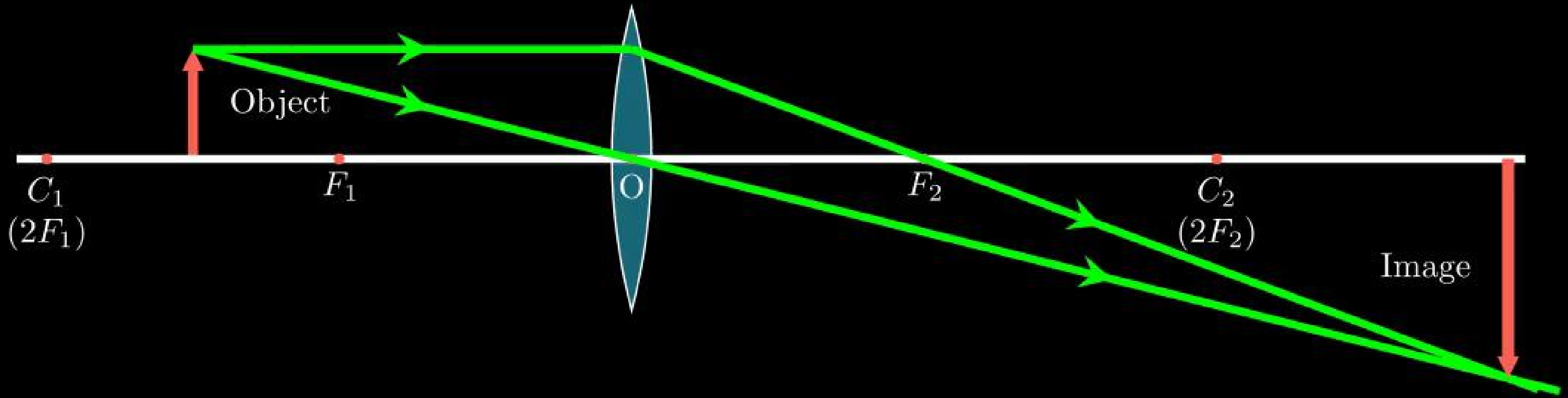
Sign convention for spherical lenses :

- For lenses, sign convention are similar to the one used for spherical mirrors. Except that all measurements are taken from the optical centre (O) of the lens.

- Focal length of Convex lens is Positive

- Focal length of Concave lens is negative





Lens Formula and Magnification

- Lens Formula: For spherical lenses, the relationship between u, v and f is given by lens formula.

$$\boxed{\frac{1}{f} = \frac{1}{v} - \frac{1}{u}}$$

- Magnification (m) : It is the ratio of height of image (h_i) to the height of the object (h_o)

$$m = \frac{h_i}{h_o}$$

- For spherical lenses the magnification m is also related to the object distance (u) and image distance (v).

$$\boxed{m = \frac{h_i}{h_o} = \frac{v}{u}}$$

Example 18 : A concave lens has focal length of 15 cm. At what distance should the object from the lens be placed so that it forms an image at 10 cm from the lens? Also, find the magnification produced by the lens.

Solution :

Example 19 : A 2.0 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 10 cm. The distance of the object from the lens is 15 cm. Find the nature, position and size of the image. Also find its magnification.

Solution :

Example 20 : An object is placed at a distance of 15 cm from a convex lens of focal length 20 cm. List four characteristics (nature, position, etc.) of the image formed by the lens. (AI2017)

Solution :

Example 21 : The image of an object formed by a lens is of magnification -1. If the distance between the object and its image is 60 cm, what is the focal length of the lens? If the object is moved 20 cm towards the lens, where would the image be formed? State reason and also draw a ray diagram in support of your answer. (AI 2016)

Solution :

Example 22 : (a) Define focal length of a spherical lens.
(b) A divergent lens has a focal length of 30 cm. At what distance should an object of height 5 cm from the optical centre of the lens be placed so that its image is formed 15 cm away from the lens? Find the size of the image also.
(c) Draw a ray diagram to show the formation of image in the above situation. (AI 2016)

Solution :

Example 23 : The image of a candle flame placed at a distance of 40 cm from a spherical lens is formed on a screen placed on the other side of the lens at a distance of 40 cm from the lens. Identify the type of lens and write its focal length. What will be the nature of the image formed if the candle flame is shifted 25 cm towards the lens? Draw a ray diagram to justify your answer. (Foreign 2014)

Solution :

Example 24 : Rishi went to a palmist to show his palm. The palmist used a special lens for this purpose.

(i) State the nature of the lens and reason foí its use.

(ii) Where should the palmist place/hold the lens so as to have a real and magnified image of an object?

(iii) If the focal length of this lens is 10 cm, the lens is held at a distance of 5 cm from the palm, use lens formula to find the position and size of the image. (2020)

Solution :

Example 25 : A student focused the image of a candle flame on a white screen using a convex lens. He noted down the position of the candle, screen and the lens as under:

Position of candle : 26 cm

Position of convex lens : 50 cm

Position of screen : 90 cm

Select the row containing the correct values as per sign convention:

Solution :

	Object Distance (u)	Image Distance (v)	Focal Length (f)
(a)	-26 cm	-50 cm	$+30$ cm
(b)	-26 cm	-40 cm	-15 cm
(c)	-24 cm	-40 cm	$+15$ cm
(d)	-24 cm	$+40$ cm	$+15$ cm

Example 26 : If the real image of a candle flame formed by a lens is three times the size of the flame and the distance between lens and image is 80 cm, at what distance should the candle be placed from the lens?

(i) -80 cm (ii) -40 cm

(iii) $-\frac{40}{3}$ cm (iv) $-\frac{-80}{3}$ cm

Solution:

Example 27 : A converging lens forms a three times magnified image of an object, which can be taken on a screen. If the focal length of the lens is 30 cm, then distance of the object from the lens is:

(i) -55 cm (ii) -50 cm

(iii) -45 cm (iv) -40 cm

Solution:

Example 29 :Analyse the following observation table showing variation of image distance (v) with object distance (u) in case of a convex lens and answer the questions that follow without doing any calculations

(a) What is the focal length of the convex lens? Give reason to justify you answer.

(b) Write the serial number of the observation which is not correct. On what basis have you arrived at this conclusion.

Select an appropriate scale and draw a ray diagram for the observation at S. No 2. Also find the approximate value of magnification.

	Object Distance (u) cm	Image Distance (v) cm
(1)	-100	$+25$
(2)	-60	$+30$
(3)	-40	$+40$
(4)	-30	$+60$
(5)	-25	$+10$
(6)	-15	$+120$

Solution :

Power of a Lens

- It is defined as the reciprocal of focal length.
- The degree of convergence or divergence (or bending) of light rays achieved by a lens is expressed in terms of its power.
- Power $= \dfrac{1}{\text{focal length}} = \dfrac{1}{f}$
- S.I. unit of Power is dioptre (D)
- 1 dioptre is the power of a lens whose focal length is 1 metre. $(1D = 1m^{-1})$
- Power of convex lens : Positive
- Power of concave lens : Negative
- Power of a lens combination : $P = P_1 + P_2 + P_3 + ...$

Example 30 : Find the power of a concave lens of focal length 20 m

Example 31 : Find the focal length of a lens of power -2.0 D. What type of lens is this?

Example 32 : A doctor has prescribed a corrective lens of power +1.5 D. Find the focal length of the lens. Is the prescribed lens diverging or converging?

Example 33 : (a) Two lens have power of +2 D and - 4 D. What is the nature and focal length of each lens?

(b) An object is kept at a distance of 100 cm for a lens of power -4 D. Calculate image distance.